

**Learning target 1.4-A, version 1**

Use the limit definition of the derivative to find a derivative function algebraically.

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Suppose that  $f(x) = 3x^2 - 5x + 4$ . Use the limit definition of the derivative to find  $f'(x)$ .

PS: The answer is  $f'(x) = 6x - 5$ .

**Learning target 1.4-A, version 2**

Use the limit definition of the derivative to find a derivative function algebraically.

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Suppose that  $g(w) = 6w^3 - 2w^2 - 9w - 2$ . Use the limit definition of the derivative to find  $g'(w)$ .

Algebra hint:  $(w + h)^3 = w^3 + 3w^2h + 3wh^2 + h^3$ .

PS: The answer is  $g'(w) = 18w^2 - 4w + 9$ .

**Learning target 1.4-A, version 3**

Use the limit definition of the derivative to find a derivative function algebraically.

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Suppose that  $p(z) = -2z^3 + 5z^2 - 4z + 4$ . Use the limit definition of the derivative to find  $p'(z)$ .

Algebra hint:  $(z+h)^3 = z^3 + 3z^2h + 3zh^2 + h^3$ .

PS: The answer is  $p'(z) = -6z^2 + 10z - 4$ .

**Learning target 1.4-A, version 4**

Use the limit definition of the derivative to find a derivative function algebraically.

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Suppose that  $f(x) = 4x^2 - 3x + 100$ . Use the limit definition of the derivative to find  $f'(x)$ .

PS: The answer is  $f'(x) = 8x - 3$ .